Recommended Should-Costing Methodology Casting + Machining and Forging + Machining

Deliverable B (brief Phase 11-B) — to be read with the Technical Audit Report (deliverable A)

> **Ground rules.** (1) Extend the existing modules; no new calculator, no rewrite of `computeUniversalStack`. (2) The CAD-only workflow stays: with no drawing every new input falls back to today's rule, stated as an assumption on the basis line. (3) The AI never sets a number. (4) No value is invented here: where a figure must come from a paywalled standard (ISO 8062-3:2007, EN 10243-1 Table 1) or a supplier, the text says "needs licensed table / source". (5) Nothing in this document is implemented; each package needs the owner's approval.

Contents

1. Manufacturing process flow
2. Engineering input requirements
3. Machining allowance methodology
4. Operation sequencing rules
5. Machine selection decision logic
6. Setup and cycle-time methodology
7. Material balance and scrap treatment
8. Complete cost calculation structure
9. Changes mapped to the existing architecture
10. Implementation packages for approval
11. Data needed from the customer

1. Manufacturing process flow

Stage	Casting + machining	Forging + machining	Output carried forward
0 Inputs	Finished drawing (+ as-cast drawing if any), finished CAD (+ raw CAD if any), volume, life, country	Same, with the forging drawing	Unified engineering model (section 2)
1 Raw shape	Finished geometry + per-face allowance (ISO 8062-3) + cored / drilled decision per hole	Finished geometry + per-face allowance + EN 10243-1 tolerance + draft + pierced / drilled decision	Raw volume and mass, per-face stock
2 Primary input	Charge = raw ÷ (1 − reject) ÷ yield poured; returns remelted, melt loss	Billet = forging + flash + wad + scale + crop / kerf	Metal bought, losses by stream
3 Primary conversion	Melt, mould / die, cores, pour, cool, knock-out, fettle	Saw, heat, preform / block / finish, trim + pierce, controlled cool	Operations with time, crew, machine
4 Secondary	Heat treat, blast, impregnate, NDT	Heat treat, descale / blast, coin, NDT	Service lines with source
5 Machining	OP10 / OP20 … from datums; passes by stock and tolerance	Same	Operations, setups, fixtures, tools
6 Inspection	In-process gauging, CMM sampling, final	Same	Inspection minutes
7 Scrap and recovery	Casting reject (remelt), machining scrap, swarf credit	Forging reject, flash / wad / crop credit, scale lost, swarf credit	Credits and uplifts
8 Cost stack	computeUniversalStack — overhead and margin once	Same	Phase 8 cost bridge

2. Engineering input requirements

Input	Source	Used for	CAD-only fallback (today's behaviour, stated)
Finished CAD (STEP)	Customer	Geometry, feature table, face map	Required (unchanged)
Raw CAD (as-cast / as-forged STEP)	Customer / foundry / forge — optional	Per-face stock = raw − finished; raw mass	Allowance rule (section 3)
Finished drawing	Customer	Machined surfaces (texture symbols), tolerances, GD&T datums, threads	Kernel face rows + all holes, flagged "assumed"

Input	Source	Used for	CAD-only fallback (today's behaviour, stated)
As-cast / as-forged drawing	Customer — optional	Raw dimensions, RMA / DCTG note, forging tolerance grade, draft, parting	ISO 8062-3 / EN 10243-1 grade by process (licensed table)
Drawing identity	Title blocks	Pair association: part number, revision, units must match; else refuse	Single drawing as today
RMA grade + DCTG grade (castings)	Drawing general note, e.g. "ISO 8062-3 – DCTG x – RMA y"	Allowance	Grade by process (needs licensed table / source: ISO 8062-3:2007 Annex tables)
EN 10243-1 grade (F / E), steel group, shape complexity S	Forging drawing / derived from mass and envelope	Forging tolerance	needs licensed table / source: EN 10243-1 Table 1
Volume, programme life, country, terms	Engineer	Batching, amortisation, rates	Required (unchanged)

CAD limitation. A finished-only CAD does not contain the raw casting or forging. The method must not present a derived raw shape as measured: every derived stock, draft and parting choice is printed as an assumption with its standard and grade. Of the four genuine pairs found (Stirling crankcase / bedplate, CH-47 gear / pinion), none has CAD, and the Stirling "raw" drawings are cast-part detail drawings rather than pure as-cast drawings.

3. Machining allowance methodology

3.1 Castings — ISO 8062-3

Quantity	Formula	Inputs	Source
Raw dimension, external feature machined both sides	$R = F + 2 \cdot RMA + CT/2$	F finished dimension; RMA for the RMA grade and the casting's LARGEST finished dimension; CT total casting tolerance for the DCTG grade and the nominal dimension	ISO 8062-3:2007 (formula); values: needs licensed table / source (ISO 8062:1994 Tables 1, 2, A.1, B.1 were read by the reference, S1; the 2007 tables were not)
Raw dimension, internal feature	$R = F - 2 \cdot RMA - CT/2$	as above	as above
One side machined (step)	$R = F + RMA + CT/2$	as above	as above
Nominal stock per side for mass	$s = RMA + CT/4$ (the cast surface lies anywhere in $\pm CT/2$ about R)	RMA, CT	Reference B convention (S1); needs licensed table / source for the values
Raw mass	$m_{raw} = \rho \cdot (V_{finished} + \Sigma_{faces} A_i \cdot s_i + \Sigma_{bores} \pi \cdot D \cdot L \cdot s + \Sigma_{holes\ drilled\text{-}from\text{-}solid} \pi/4 \cdot D^2 \cdot L)$	Measured areas (kernel), per-face s_i	Arithmetic

3.2 Forgings — EN 10243-1 + machining allowance

Quantity	Formula	Inputs	Source
Tolerance	T (+ / −) from mass, steel group M1 / M2, shape complexity S = m_forging / m_envelope, grade F or E, dimension	Raw mass (iterate), envelope, grade	EN 10243-1:1999 method (S6); Table 1 values: needs licensed table / source
Nominal stock per side	$s = a_{finish} + \lvert T_- \rvert$ (minus tolerance) ; scale pits ≤ ½ a_{finish}	a_{finish} per surface	a_{finish}: IS 3469 (Part I):1974 Table 1.5 (already in forging.ts) or the customer standard; EN 10243-1 §5.2.4 for scale pits
Draft metal	$V = \Sigma A \cdot h \cdot \tan \alpha / 2$ per zero-draft face	Draft angle by surface	IS 3469 Table 1.4 (already in forging.ts:318-361)

Quantity	Formula	Inputs	Source
Pierced v drilled	Bore pierced when D ≥ D_pierce and web allows; else drilled from solid	D, web thickness	Forge practice — needs licensed table / source

3.3 Roughing / finishing split and checks

Rule	Formula / logic	Source
Finishing stock	s_finish = the insert / tool's recommended finishing ap for the metal (catalogue row), capped by s	Maker cutting data (cutting-catalogue.ts rows; S2–S5)
Roughing passes	n_rough = ceil((s − s_finish) / ap_rough,max)	Maker ap limits (catalogue)
Semi-finish / finish / fine-bore / ream / grind / hone	Selected by the feature's IT grade and Ra: tighter than the finishing capability → add the next process	Process-capability table — needs licensed table / source (ISO 286 grades v process charts)
Insufficient stock flag	flag if s < s_finish or s < scale / skin depth	EN 10243-1 §5.2.4 (scale pits); foundry skin — needs licensed table / source
Conflict flag	flag if a raw-drawing dimension implies s ≤ 0 or s > 2 × the standard value	Arithmetic

4. Operation sequencing rules

#	Rule	Input
S1	OP10 locates on cast / forged datum targets (ISO 8062 datum system, or the drawing's target points) and machines the features that become datums A / B / C first.	Drawing datums (gdt.datum, already extracted)
S2	Features related by a tight position / runout / concentricity tolerance to the same datum go in the same clamping.	gdt.frame (already extracted)
S3	A turned axis with bearing seats is turned between centres or chucked in its own OP, and bores that must run true to it are bored from it (e.g. king-pin bores after the spindle).	Turning signature + tolerance frames
S4	Roughing of all features precedes finishing of datum-related features; heavy stock removal before the final clamping.	Per-face stock (section 3)
S5	Each OP lists: features, operation, tool, workholding, orientation, machine, axes, cycle, tolerance, inspection.	Brief Phase 4 table
S6	Inspection: in-process gauging for every toleranced feature tighter than a threshold; CMM sampling per the control plan.	Customer control plan — needs licensed table / source

5. Machine selection decision logic

Step 1 — feasibility (new). For each candidate class, compute the minimum number of clampings that reaches every SIGNED approach direction of the machined features, with the clamped face excluded; reject a class whose envelope, table load or capability (tightest IT grade) the part exceeds.

Candidate	Reach per clamping	Clampings (minimum set cover)	Typical trade-off
3-axis VMC, multiple setups	One direction	= number of distinct signed directions	Cheapest machine £/h; most handling, fixtures and datum transfers
4-axis (VMC + rotary, or HMC with B axis)	All directions ⊥ the rotary axis	Set cover by rotary axes; +1 for the clamped face if machined	Fewer setups; tombstone fixtures
5-axis trunnion	Hemisphere above the clamped face	≥ 2 if opposite faces are machined	Highest £/h; fewest datum transfers; compound angles
Twin-pallet HMC cell	As 4-axis; load masked by the cut	As 4-axis	Volume production; hydraulic fixtures
CNC lathe / mill-turn	Revolved features about one axis	2 chuckings for op 10 / op 20	Turned parts; holes off-axis need a mill or live tooling

Step 2 — cost. Rank only the feasible candidates by machine + operator + fixtures + setup amortisation per part (the existing `optimiseMachiningRouting` arithmetic), and print each rejected candidate with the reason. Fewer setups is never assumed cheaper.

Evidence that step 1 is missing today: probe P3 — the bracket needs 7 approach directions, PRCR002 8, the knuckle 9, Part1 11; none can be reached in one clamping on a single rotary axis, yet each was costed in 1.

6. Setup and cycle-time methodology

Element	Formula	Inputs and source
Cutting time per feature	$t_{cut} = \Sigma_{passes} L_{pass} / v_f$; $v_f = n \cdot z \cdot f_z$ (milling) or $n \cdot f$ (drilling, boring, turning); $n = 1000 \cdot v_c / (\pi \cdot D)$	Maker data by metal and operation (cutting-catalogue.ts, sourced); passes from section 3.3
Non-cutting time per OPERATION (replaces the per-feature allowance)	$t_{noncut} = N_{tool} \cdot t_{tc} + N_{approach} \cdot t_{app} + N_{index} \cdot t_{index} + t_{probe} + t_{pallet}$	t_{tc} 2.2 s (Okuma / DMG, sourced); t_{index}, t_{probe}, approach / rapid from the machine maker — needs licensed table / source
Cycle per operation	$t_{op} = t_{cut} + t_{noncut}$	—
Machine time per part	$\Sigma t_{op} \div OEE$	OEE: plant data — needs licensed table / source (today 0.8, unsourced)
Labour per part	$\Sigma t_{op} \times$ crew ÷ efficiency + handling not masked by the cut	Crew and grade by cell type — needs licensed table / source
Initial setup	Fixture change + tool offsets + first-off approval per clamping, hours per batch	Plant time study — needs licensed table / source
Setup per part	Σ setups ÷ batch; batch from the customer call-off or one month of demand	Call-off pattern
Forge-line and die-change set-up	$h_{change} \times$ (press + trim + heater + crew £/h) ÷ forgings between re-sinks	Die life between re-sinks — needs licensed table / source
Tool wear	£/min × t_{cut} ONLY (never t_{noncut})	tool-wear £/min (rate book) or tool life per edge — needs licensed table / source

> Why: the tool's fixed per-feature allowances are 28–76 % of its machining minutes on the five validation parts (probe P10) and add £0.70–£2.58 a part before overhead and margin (£0.62–£1.59 a part net of the reference's own per-operation non-cut); tool wear is charged on them as well (CF-19, CF-20). The reference's per-operation non-cut model (tool changes 3 s, approaches 2 s, probe 12 s, B-index 2 s, pallet change) is the structure recommended here; its values are assumptions to be replaced by machine data.

7. Material balance and scrap treatment

Stream	Casting + machining	Forging + machining	Recover-able?	Credit
Input	Charge (ingot / pig + scrap + alloys) = raw + melt loss	Bar = billet + kerf + crop share	—	—
Gating, feeders, rejected castings	Poured − raw; remelted in-house	—	Internal	None (remelted; only the melt loss is bought)
Melt loss / dross	Loss fraction × poured returns + new metal	—	No	0
Flash	—	Flash-line perimeter × land width × flash thickness × ρ (the trim-press geometry already computed in forging.ts flashGeometry)	Yes	Steel scrap £/kg
Pierce wad	—	$\pi/4 \cdot d^2 \cdot t_{web} \times \rho$	Yes	Steel scrap £/kg
Scale	—	Scale % × billet by heating method — needs licensed table / source	No	0

Stream	Casting + machining	Forging + machining	Recover-able?	Credit
Kerf and crop ends	—	π/4·D²·kerf × ρ; crop ÷ billets per bar	Yes	Steel scrap £/kg
Machining swarf	Raw − finished	Raw − finished	Yes	Turnings / borings £/kg — needs licensed table / source
Scrapped castings / forgings	Reject % (remelt)	Reject %	Yes	Scrap £/kg
Scrapped machined parts	Machining scrap % × (raw part + machining done)	Same	Metal only	Scrap £/kg on the metal

No double counting: (1) a loss is either remelted (casting) or bought-and-credited (forging), never both; (2) yield is an OUTPUT (finished ÷ input) checked against a band, never an input multiplied on top of itemised losses; (3) reject uplift applies to the operations a scrapped part has passed, once; (4) services priced by a supplier carry a handling %, not the costing's overhead and margin again (core.ts boughtIn already supports this); (5) the balance must close to 0.001 kg and print every stream.

8. Complete cost calculation structure (brief Phase 8)

#	Line	Calculation	Today's bucket
1	Raw material input	Σ input kg × £/kg	rawMaterial
2	Primary conversion (casting / forging)	Σ primary operations: rate × time ÷ OEE + labour	process + labour
3	Secondary processes	Heat treat, blast / descale, impregnation, coining — build-up or sourced toll price + handling	rawMaterial (consumables) → separate line
4	Machining by operation	Per OP: machine + labour (section 6)	process + labour
5	Setup and fixtures	Setup per part + fixtures ÷ programme	process / tooling
6	Tooling amortisation	Dies, patterns, core boxes: first set + wear sets ÷ programme parts	tooling
7	Labour	Shown per operation	labour
8	Machine	Shown per operation	process
9	Inspection and quality	NDT, gauging, CMM sampling	rawMaterial (NDT) → separate line
10	Scrap, yield, recovery	Uplifts − credits (section 7)	inside material / operations → separate line
11	Overheads and margin	12 % of factory base, 8 % of subtotal, once (rate-book oh-default)	overhead, margin

The eight-bucket headline stays (it is the engine's contract); the eleven lines are a report view over the same trace, which must re-add to the headline to £0.0001.

9. Changes mapped to the existing architecture

Change	Existing module / function it extends	Acceptance criteria	Test evidence required
Machined-face set from the drawing / raw CAD; draft excludes a face; no top-8 cap	cad-geometry-engine.py _extract_machining_features (:330-449); feature-machining.ts defaultInclude; dfm-geometry/machined-faces.ts machinedFaceSet; engineering-data packs (bind finish.roughness, gdt)	Modelled yoke: 4 eye faces + 2 bores; CAD-only default unchanged and labelled	Unit test on the yoke STEP; real-parts baseline diff reviewed; ≥ 3 genuine drawing pairs checked by an engineer (≥ 95 % face precision / recall)
Per-face allowance (ISO 8062-3 / EN 10243-1)	machining-time.ts castMachiningStockMm, nearNetStockCm3, asCastWeight; forging.ts machiningStockMm	Raw mass within ±3 % of a foundry / forge weight on ≥ 3 parts	Licensed tables loaded as data with source; tests per grade and size band

Change	Existing module / function it extends	Acceptance criteria	Test evidence required
Passes by stock and tolerance	machining-time.ts nearNetMachiningTime; feature-machining.ts featureTime; cutting-catalogue.ts	Minutes rise with stock; H7 bore gets a finish operation	reference-part.test.ts (£24.79) still reconciles; new unit tests
Non-cut time per operation; tool wear on cut time	feature-machining.ts ALLOWANCE_MIN; machining.ts buildOperationPlan, cuttingMinutesOf; machining-time.ts toolWearPerPart	Tool wear unchanged when non-cut changes; non-cut within ±20 % of a cycle sheet	2–3 supplier cycle sheets
Signed approach directions on machined features; clamping set cover	cad-geometry-engine.py _compute_setup_count; routing-optimiser.ts optimiseMachiningRouting	Bracket, PRCR002, knuckle, Part1 ≥ 2 clampings; 1-face part stays 1	Probe P3 turned into a unit test
OP10/OP20 grouping from datums	machining.ts buildOperationPlan	Hub flange bore and OD in one lathe set-up; one load per chucking	Route card comparison on ≥ 1 part
Forging balance by stream; credits by stream	forging.ts advise / yieldFractionFor / flashGeometry; modules/forging.ts computeForgingDrivers; core.ts scrap credit	Balance prints every term, closes to 0.001 kg; billet ±5 % of a forger's cut weight	1 genuine forging with cut weight
Machining scrap on near-net routes	cost-input-rules/commodities/cast-and-machine.ts DROPPED_MACHINING_RULES; modules/cast-and-machine.ts	A 2 % rate raises cost by the computed amount	Unit test
Core, fettling, blast, heat-treat, inspection build-ups	casting.ts rules coreCostPerPart, fettling, shot blast, heat treat; service-rates-data.ts	Each line prints its source or stays labelled	Supplier quotes / time studies
Cost bridge view	src/export/* report models (executive-summary.ts buildCoreReport)	Phase 8 lines re-add to the headline	Report test
Bevel gear refusal (or capability)	cost-input-rules/derive/commodity.ts; gear rules	A bevel gear is refused or costed on a bevel route	CH-47 models

10. Implementation packages for approval

Expected effects are the directions and sizes measured by the audit probes (UK; P4 / P6 at 50k/yr, P10 / A v B at 200k/yr × 5 yr). They are not predictions of the corrected cost; each must be re-measured after implementation.

Package	Gaps	Expected direction and size (from probes)	Effort	Risk	Data needed from the customer
PKG-1 Machined surfaces, allowance and the raw-to-finished model	CF-01, CF-02, CF-03, CF-07	Down where the face set is over-selected (yoke −39.5 % when only the eye faces are machined; knuckle −£1.35 without the lathe "spindle"); up or down ≈ ±3 % per ±1.5 mm of stock (bracket −£0.31, PRCR002 −£0.88 for −1.5 mm). Machining time then moves with stock (it does not today).	L	Medium — the CAD-only default must stay; a wrong face map moves cost both ways	JLR's licensed ISO 8062-3:2007 and EN 10243-1 tables; drawings with the RMA / DCTG general note and surface-texture symbols; raw (as-cast / as-forged) CAD where it exists
PKG-2 Cycle-time model: non-cut time per operation, tool wear on cutting time, passes by stock and tolerance	CF-19, CF-20, CF-06, CF-16, CF-22	Allowances alone are £1.42 / £2.58 / £2.56 / £1.68 / £0.70 a part before overhead and margin on bracket / Part1 / PRCR002 / knuckle / hub; tool wear on them up to £0.63. Net of B's own per-operation non-cut, A's non-cutting time is £0.62–£1.59 a part higher. Adding rough + finish passes raises cutting time where stock and tolerance need it (knuckle HMC: A 2.9 v B 3.8 min). Expected net direction: down.	M	Medium — changes every machined part; the reference part test (£24.79) must still reconcile	Supplier cycle sheets for 2–3 parts; machine rapid traverse, B-axis index and probing times from the maker
PKG-3 Setups, operation sequence with datums, machine-selection feasibility	CF-05, CF-08, CF-15	Up: 2 HMC clampings add +3.6 % / +2.5 % / +5.0 % / +3.3 % (bracket / PRCR002 / knuckle / Part1, 50k); hub flange second lathe load ≈ £0.46 before overhead and margin.	M–L	Medium — routing ranking can flip once setups are real	Drawing datums (already read by the drawing reader); fixture concepts or supplier route cards; machine envelopes and table loads
PKG-4 Forging material balance and scrap / recovery streams	CF-04, CF-14, CF-09	Hub flange metal +£0.85 v B (billet 4.72 v 3.95 kg); yield band ±: knuckle +8.7 % / −6.0 %; swarf credit −£0.04 to −£0.29; machining scrap ≈ +2 %.	M	Low–Medium — the forging arithmetic is isolated in forging.ts / modules/forging.ts	Forge-shop cut weights and flash / wad data for 1–2 forgings; a sourced scale % by heating method; turnings / borings scrap prices; machining scrap rates
PKG-5 Foundry and secondary-operation build-ups (cores, yield, fettling, blast, heat treatment, inspection)	CF-10, CF-11, CF-12, CF-13	Cores: PRCR002 +£2.77 and knuckle +£0.77 above B; fettling +£0.49 to +£0.51 on the aluminium parts; inspection −£0.33 to −£0.55 (B carries gauging / CMM A lacks); casting yield 0.40 / 0.75 moves PRCR002 +7.1 % / −1.9 %.	M	Low — each line is a separate rule	Core-shop data and binder price; heat-treat toll quotes (T6, normalise); blast / fettle times from a foundry; the customer's control plan (inspection frequency)
PKG-6 Cost bridge and reporting	CF-17, CF-18	No change to totals; the Phase 8 lines (raw material, primary conversion, secondary, machining by operation, inspection, scrap) become visible.	S	Low	None
PKG-7 (separate decision) Bevel / spiral-bevel gear capability (or an explicit refusal)	CF-21, CF-22	Not quantifiable in £: recorded / tool labour hours 14× (forged route) and 61× (conventional route), era and scope inside the ratios.	L build / S refusal	High for a capability build	Bevel-gear drawings with data blocks; Gleason-type cycle data from a gear supplier

Suggested order: PKG-6 (no cost change, makes the others visible) → PKG-1 and PKG-2 together (they interact: stock drives passes) → PKG-3 → PKG-4 → PKG-5. PKG-7 is a separate scope decision. After PKG-1 to PKG-4, re-run the five validation parts and at least three parts with real prices before any accuracy statement.

11. Data needed from the customer

Data	Why	Package
Licensed ISO 8062-3:2007 (and :2023 if adopted) and EN 10243-1 copies	RMA / DCTG and forging tolerance values cannot be invented or taken from an unlicensed copy	PKG-1
Raw (as-cast / as-forged) drawings and CAD for ≥ 3 production parts	Measure stock instead of deriving it; validate the face map	PKG-1
Supplier quotes or purchase prices with volume and terms	The only way to measure accuracy (scripts/actuals/cad-actuals-input.csv)	All
Supplier cycle sheets / route cards for 2–3 machined castings or forgings	Non-cut time, setups, OP structure	PKG-2, PKG-3
Machine data: rapid traverse, B-axis index, probing cycle, table load	Non-cut model and feasibility	PKG-2, PKG-3
Forge-shop cut weights, flash and wad data; scale % by heating method	Forging balance	PKG-4
Core-shop data, binder price; heat-treat toll prices; control plans	Secondary and inspection build-ups	PKG-5